PES UNIVERSITY

Department of Computer Science & Engineering

Microprocessor & Computer Architecture
UE24CS251B

MPCA Lab Programs - 6

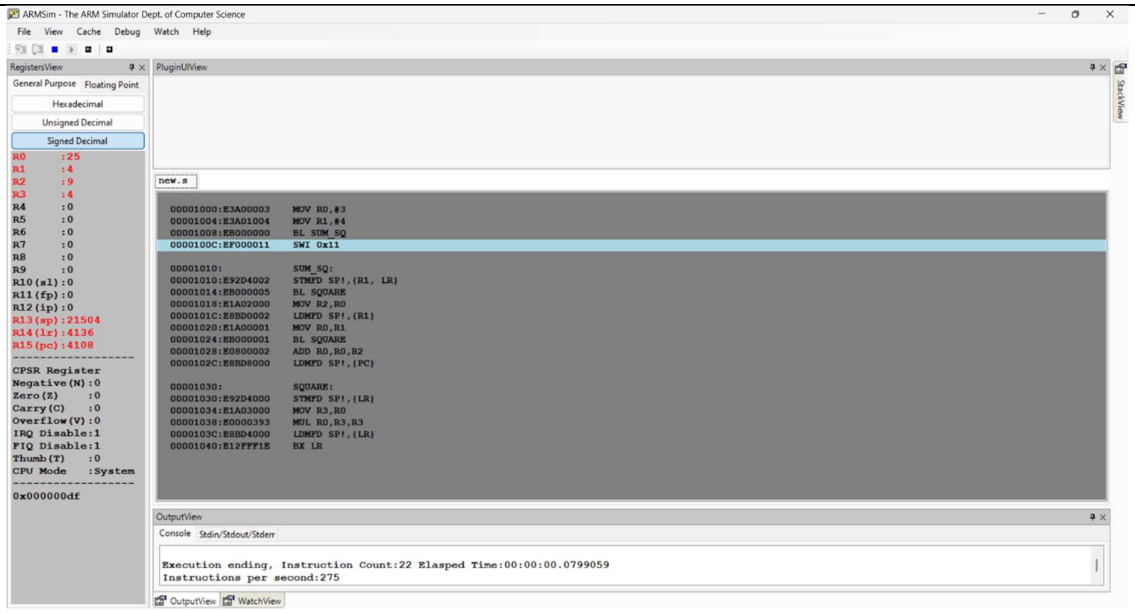
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1 **Write an ARM7TDMI assembly program using nested procedure calls to compute the sum of squares of two numbers. The main program calls a procedure SUM_SQ, which in turn calls another procedure SQUARE to compute the square of a number.**

```
MOV R0,#3
MOV R1,#4
BL SUM_SQ
SWI 0x11

SUM_SQ:
STMFD SP!,{R1, LR}
BL SQUARE
MOV R2,R0
LDMFD SP!,{R1}
MOV R0,R1
BL SQUARE
ADD R0,R0,R2
LDMFD SP!,{PC}

SQUARE:
STMFD SP!,{LR}
MOV R3,R0
MUL R0,R3,R3
LDMFD SP!,{LR}
BX LR
```



2 **Write an ARM7TDMI assembly language program to multiply two matrices of suitable order and store the result in a separate memory location.**

```
LDR R0,=A
LDR R1,=B
LDR R2,=C

LDR R3,[R0]
LDR R4,[R0,#4]
LDR R5,[R0,#8]
LDR R6,[R0,#12]

LDR R7,[R1]
LDR R8,[R1,#4]
LDR R9,[R1,#8]
LDR R10,[R1,#12]

MUL R11,R3,R7
MUL R12,R4,R9
ADD R11,R11,R12
STR R11,[R2]

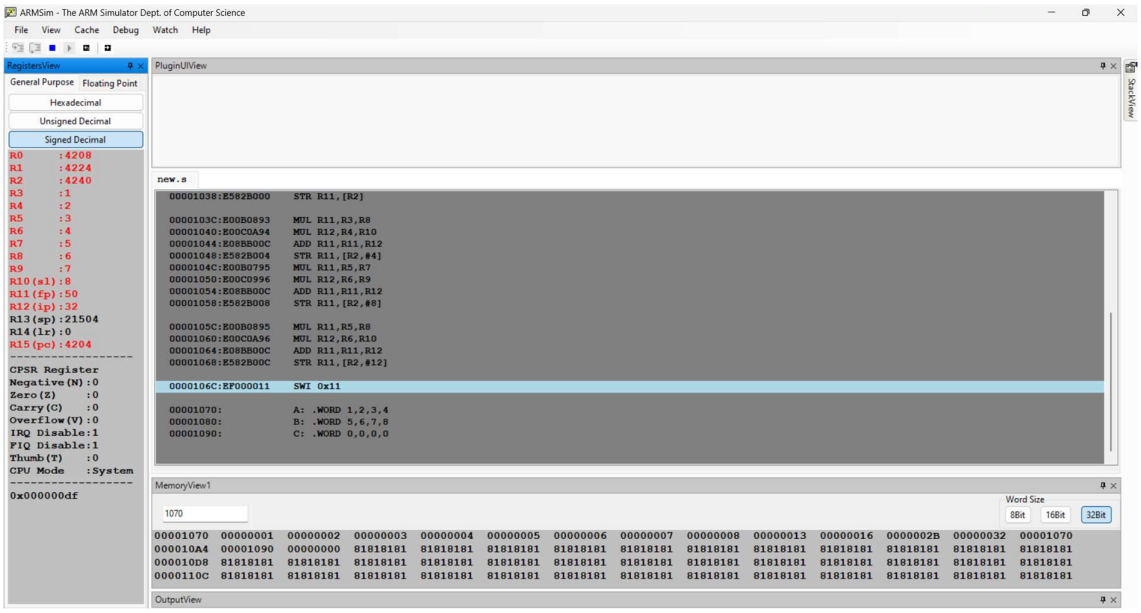
MUL R11,R3,R8
MUL R12,R4,R10
ADD R11,R11,R12
STR R11,[R2,#4]
MUL R11,R5,R7
MUL R12,R6,R9
ADD R11,R11,R12
STR R11,[R2,#8]

MUL R11,R5,R8
MUL R12,R6,R10
```

ADD R11,R11,R12
STR R11,[R2,#12]

SWI 0x11

A: .WORD 1,2,3,4
B: .WORD 5,6,7,8
C: .WORD 0,0,0,0



3 **Write an ARM7TDMI assembly language program to generate the first N terms of an arithmetic progression and store them in memory.**

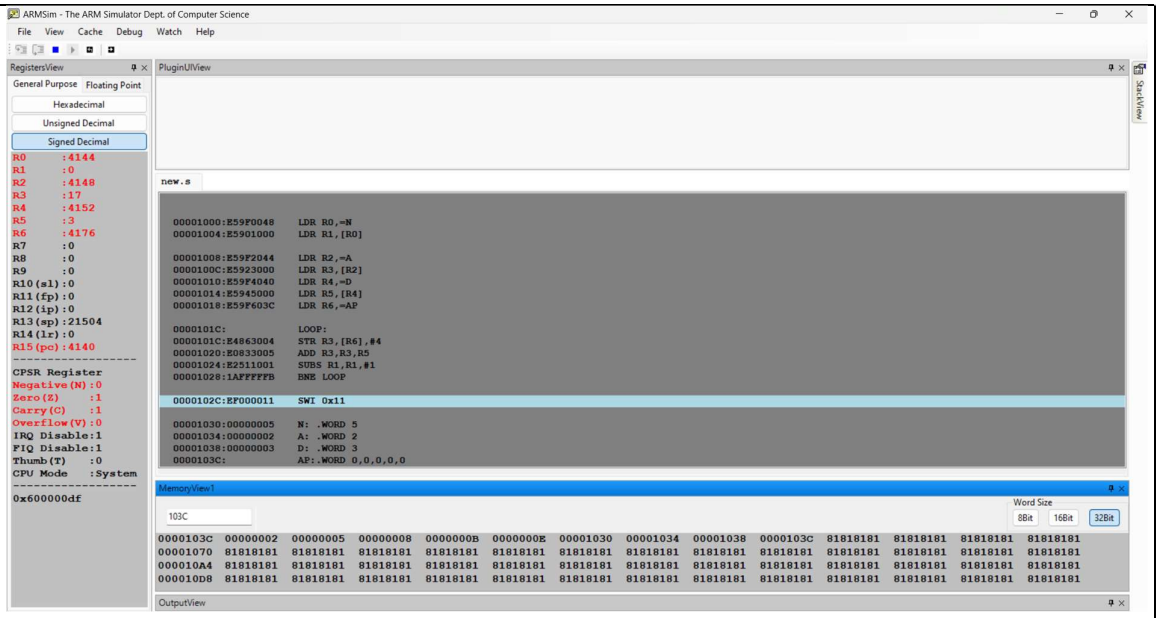
LDR R0,=N
LDR R1,[R0]

LDR R2,=A
LDR R3,[R2]
LDR R4,=D
LDR R5,[R4]
LDR R6,=AP

LOOP:
STR R3,[R6],#4
ADD R3,R3,R5
SUBS R1,R1,#1
BNE LOOP

SWI 0x11

N: .WORD 5
A: .WORD 2
D: .WORD 3
AP:.WORD 0,0,0,0,0



4 **Write an ARM7TDMI assembly language program to concatenate two strings and store the result and displays the string.**

```
LDR R0,=STR1
LDR R1,=STR2
LDR R2,=RES

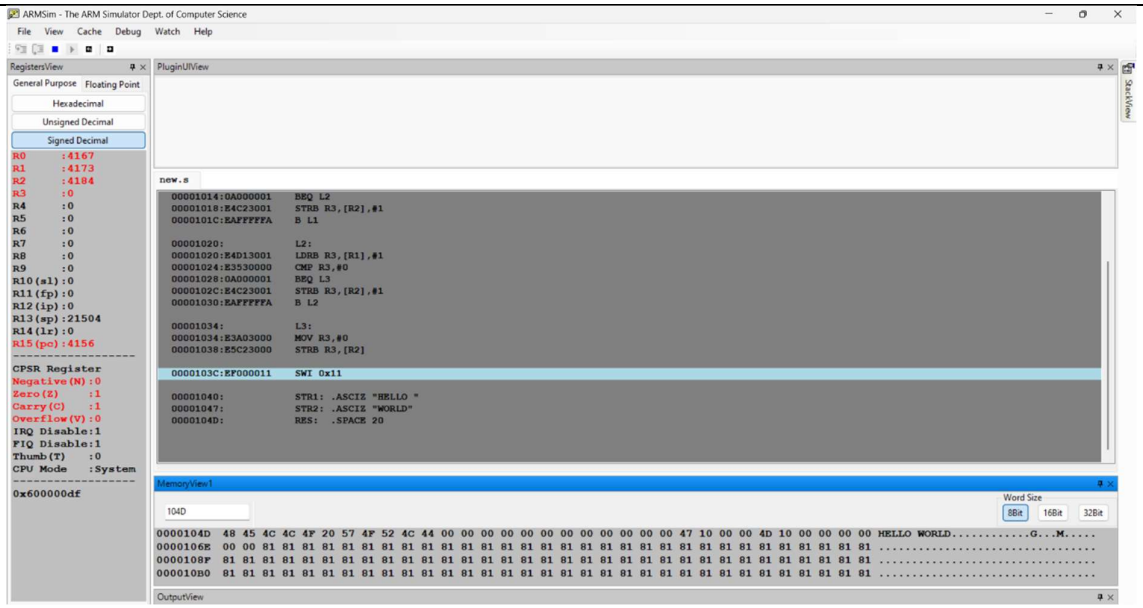
L1:
LDRB R3,[R0],#1
CMP R3,#0
BEQ L2
STRB R3,[R2],#1
B L1

L2:
LDRB R3,[R1],#1
CMP R3,#0
BEQ L3
STRB R3,[R2],#1
B L2

L3:
MOV R3,#0
STRB R3,[R2]

SWI 0x11

STR1: .ASCIZ "HELLO "
STR2: .ASCIZ "WORLD"
RES:  .SPACE 20
```



5 **ASSIGNMENT QUESTIONS:**

i) Write a program to swap the first and last character of a given string.

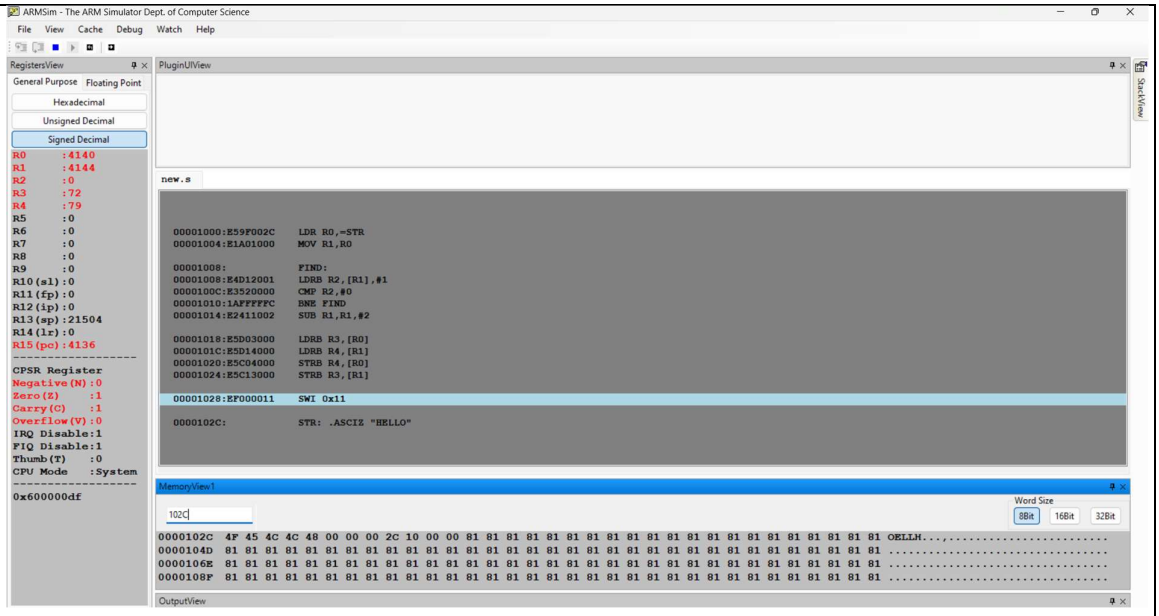
```
LDR R0,=STR
MOV R1,R0

FIND:
LDRB R2,[R1],#1
CMP R2,#0
BNE FIND
SUB R1,R1,#2

LDRB R3,[R0]
LDRB R4,[R1]
STRB R4,[R0]
STRB R3,[R1]

SWI 0x11

STR: .ASCIZ "HELLO"
```



ii) Given a c Code convert it in its equivalent Arm Code : z = (a << 2) |(b & 15);

```
LDR R0,=A
LDR R1,[R0]

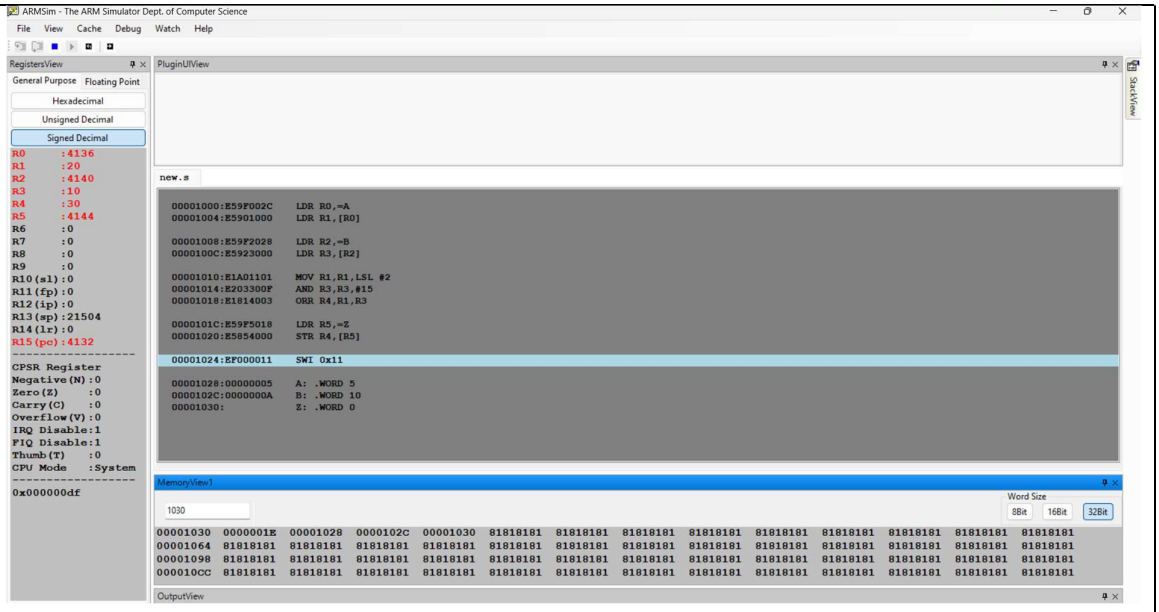
LDR R2,=B
LDR R3,[R2]

MOV R1,R1,LSL #2
AND R3,R3,#15
ORR R4,R1,R3

LDR R5,=Z
STR R4,[R5]

SWI 0x11

A: .WORD 5
B: .WORD 10
Z: .WORD 0
```



6 **Write an ARM7TDMI assembly language program in which the main program calls a procedure STRING_PROC. The procedure STRING_PROC calls separate procedures LENGTH to find the length of a string and REVERSE to reverse the string. The reversed string must be stored in a separate memory location.**

```
LDR R0,=STR
LDR R1,=REV
BL STRING_PROC
SWI 0x11

STRING_PROC:
STMFD SP!,{R4,R5,LR}
MOV R4,R0
MOV R5,R1
BL LENGTH
MOV R2,R0
MOV R0,R4
MOV R1,R5
BL REVERSE
LDMFD SP!,{R4,R5,PC}

LENGTH:
MOV R3,#0
L1:
LDRB R2,[R0],#1
CMP R2,#0
BEQ L2
ADD R3,R3,#1
B L1
L2:
MOV R0,R3
BX LR
```

```
REVERSE:
CMP R2,#0
BXEQ LR
ADD R0,R0,R2
SUB R0,R0,#1
L3:
CMP R2,#0
BEQ L4
LDRB R3,[R0],#-1
STRB R3,[R1],#1
SUB R2,R2,#1
B L3
L4:
MOV R3,#0
STRB R3,[R1]
BX LR

STR: .ASCIZ "HELLO"
REV: .SPACE 20
```

